Quiz 2 Solutions — April 28, 2026

CPSC 406 — Algorithm Analysis, Section 2
Chapman University · Spring 2026
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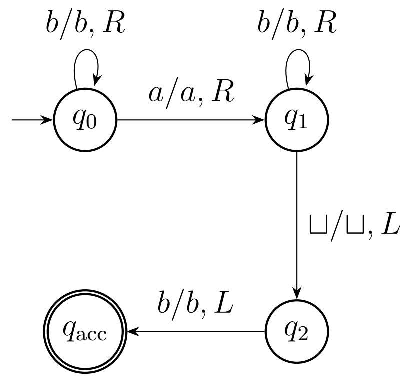
1. (2 points) (a) Consider the following Turing machine over input alphabet $\Sigma = \{a, b\}$. The blank symbol is $\sqcup$. All omitted transitions go to a rejecting halt state.



Which of the following words is accepted by the machine? Tick exactly one.

abb   *aab*   *aabb*   All of the above   None of the above

Solution. The accepted word is $\boxed{abb}$.

The machine may read some initial b's while staying in q_0, then reads one a, and then may read more b's while staying in q_1. It accepts only if the symbol immediately to the left of the final blank is b. Among the listed options, only *abb* has this form.

The other choices are wrong as follows. The word *aab* rejects because after reading the first a, state q_1 has no shown transition on a second a. The word *aabb* rejects for the same reason. Therefore "All of the above" is wrong, and "None of the above" is wrong because *abb* is accepted.

(b) Let

$$f(n) = 4n^3 + 8n \log n + 1.$$

Which of the following Big-O classes contain $f(n)$? Tick exactly one.

$\mathcal{O}(n^3)$ $\mathcal{O}(n^3 \log n)$ $\mathcal{O}(\exp(n))$ All of the above   None of the above

Solution. The correct choice is $\boxed{\text{All of the above}}$.

Since $f(n) = 4n^3+8n \log n+1$, the leading term is cubic, so $f(n) \in \mathcal{O}(n^3)$. Therefore it is also in $\mathcal{O}(n^3 \log n)$ and in $\mathcal{O}(\exp(n))$. Each listed class is correct, so choosing only one of the three individual classes is incomplete, and "None of the above" is wrong.

2. (2 points) Show that the following formula, given in CNF, is unsatisfiable using resolution:

$$\varphi = (p \lor q) \land (\neg p \lor r \lor s) \land (\neg q \lor r) \land (\neg r \lor t) \land (\neg s \lor t) \land \neg t \land \neg q.$$

Write the clauses and derive the empty clause $\Box$.

Solution.

Let the clauses be

$$\begin{aligned}
&C_1 = p \lor q, && C_2 = \neg p \lor r \lor s,\\
&C_3 = \neg q \lor r, && C_4 = \neg r \lor t,\\
&C_5 = \neg s \lor t, && C_6 = \neg t,\\
&C_7 = \neg q.
\end{aligned}$$

A resolution refutation is:

$$\begin{array}{rcll}
C_1, C_7 & \vdash & p, & \text{resolving on } q,\\
C_2, p & \vdash & r \lor s, & \text{resolving on } p,\\
C_4, C_6 & \vdash & \neg r, & \text{resolving on } t,\\
r \lor s, \neg r & \vdash & s, & \text{resolving on } r,\\
C_5, s & \vdash & t, & \text{resolving on } s,\\
t, C_6 & \vdash & \Box, & \text{resolving on } t.
\end{array}$$

Thus the formula is unsatisfiable.

Common resolution mistakes. Each resolution step should use exactly two clauses and resolve on exactly one variable. Do not cancel two variables at once, do not drop unrelated literals, and do not resolve two literals with the same sign. Also, every derived clause should come from earlier clauses or original clauses, and the refutation is complete only when the empty clause $\Box$ is derived.